Branching Logic

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected branch-and-boundary record.

Explain the chain, trace paths, repair two independent if statements and build the new tile sorter before retesting its boundary.

Keep one habit: Keep the rule, code version, checked conditions, actual result and regression cases together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Decompose a Goal

MAIN LESSON · SECTION 12

W1 · Explain the main chain and final print

> Use the supplied complete duration script.

Your evidence record

- Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented **branch**es.

```
minutes = 12
if minutes < 0:
    label = "Invalid"
elif minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Record checked and skipped conditions

> Keep expected labels separate from actual fresh runs.

Your evidence record

- For -1, 12 and 13, record checked **condition**s, skipped **condition**s and the final label. Run each as a fresh script after predicting.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair an overwritten label

> Use the supplied faulty two-if script.

Your evidence record

- Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

```
minutes = -1
if minutes < 0:
    label = "Invalid"
if minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Build a new sorter independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a new sorter independently

MAIN LESSON · SECTION 16

W5 · Retest the independent six-tile boundary

> After W4, distinguish the changed equality case from earlier category tests.

Your evidence record

- A repair changes the small-count **condition** from < 6 to <= 6. Name the case that exposes the original error and two different kinds of earlier cases to rerun.

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Branch when a reader has a hard constraint such as language, format, or maximum length.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the tested path for **Branch**ing and **boundary** tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: “Every **condition** in the chain always runs”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long.
Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the **boundary**.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

> **My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

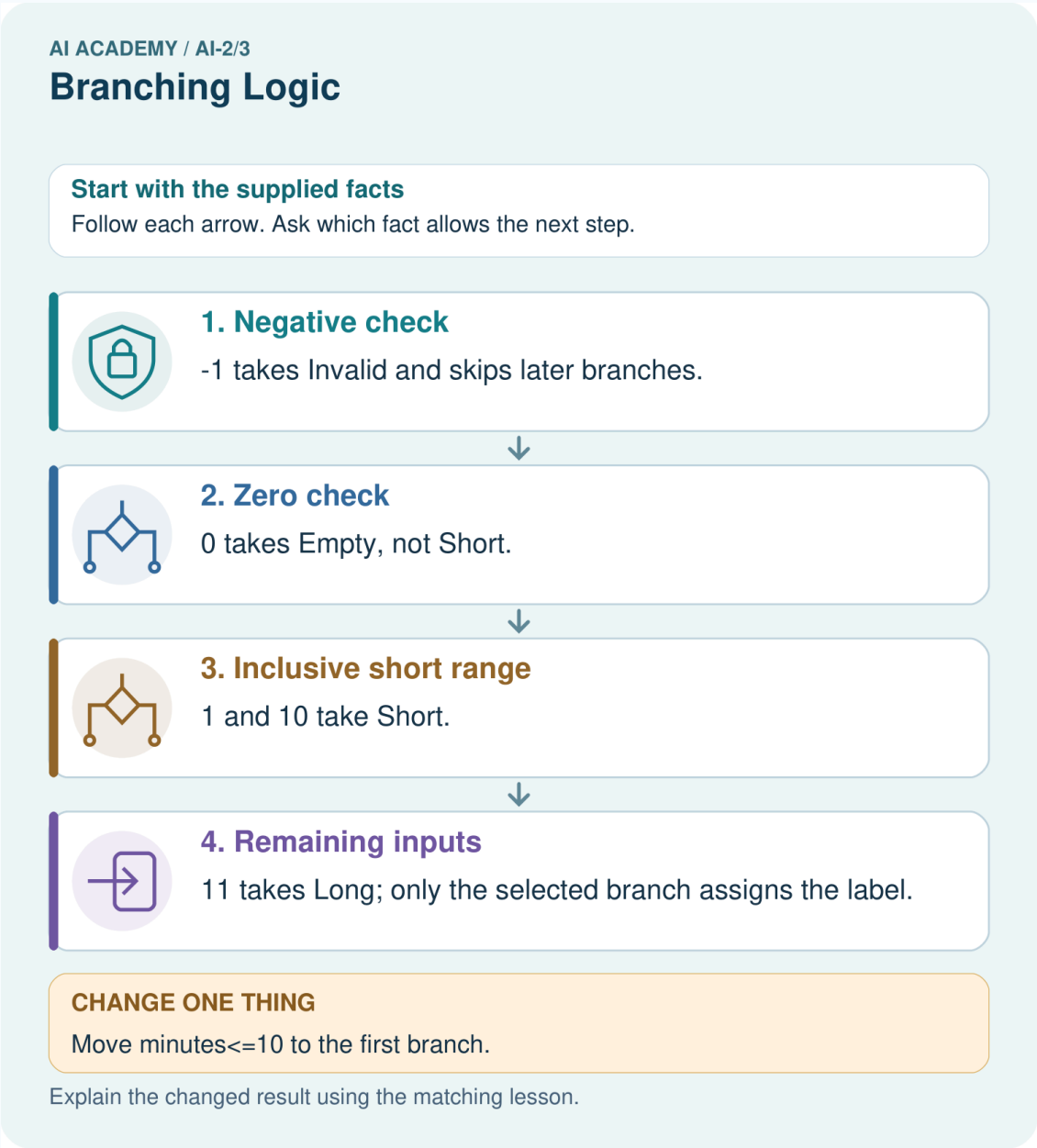
> Keep the explanation and the support you used.

Use this when helpful

A fictional reading-time labeler uses one Python if/el**if/else** chain.

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as minutes<0, minutes==0, minutes<=10, otherwise. Print the assigned label once.



Follow the labelled picture with the matching explanation. Open the vector diagram

Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional reading-time labeler uses one Python if/el**if/else** chain.
- Which labelled step matches this detail? 11 takes Long; only the selected **branch** assigns the label.
- 1. Negative check
- 2. Zero check

- 3. Inclusive short range
- 4. Remaining **input**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Remaining **input**s 11 takes Long; only the selected **branch** assigns the label.

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Changed-case question

- Changed-case practice: Move minutes<=10 to the first **branch**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. –1 and 0 wrongly receive Short.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: –1 and 0 wrongly receive Short. Both satisfy the earlier comparison before their intended checks are reached.

Repair and save your work

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

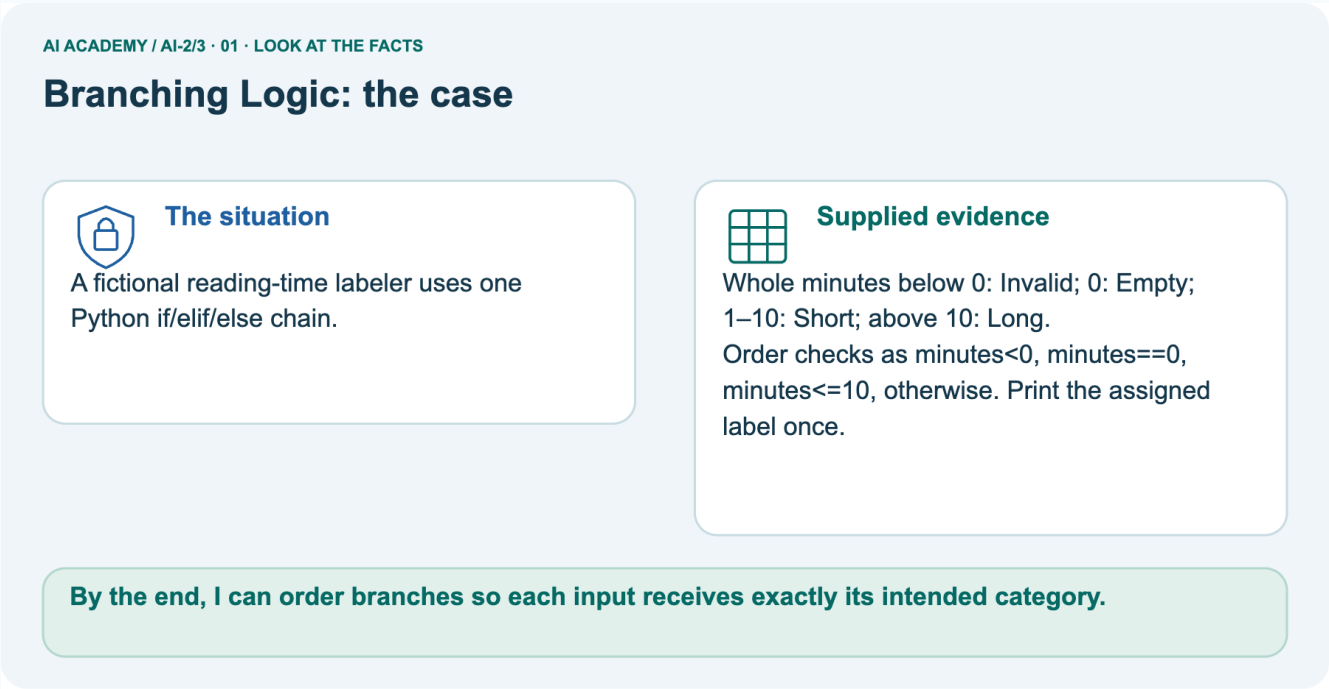
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Branching Logic: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

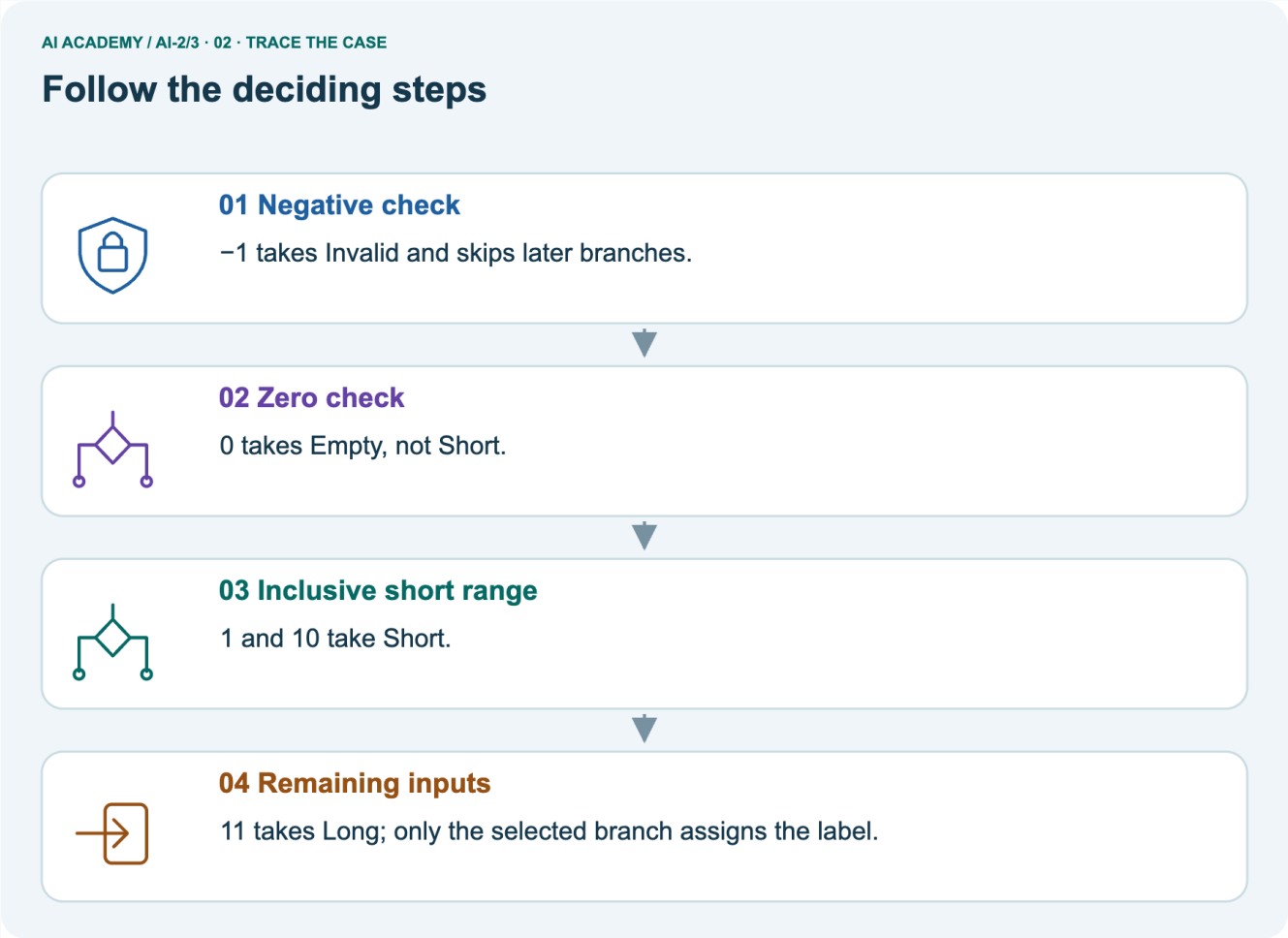
A fictional reading-time labeler uses one Python if/el**if/else** chain. Supplied facts: Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long. **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise.

Print the assigned label once.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Negative check: −1 takes Invalid and skips later **branch**es. → Zero check: 0 takes Empty, not Short. → Inclusive short range: 1 and 10 take Short. → Remaining **input**s: 11 takes Long; only the selected **branch** assigns the label.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One chain chooses one label

AI ACADEMY / AI-2/3 · 03 · INSPECT THE LESSON MODEL

One chain chooses one label

Input	Checks in order	Selected label
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

Later branches in this chain are skipped after a condition succeeds.

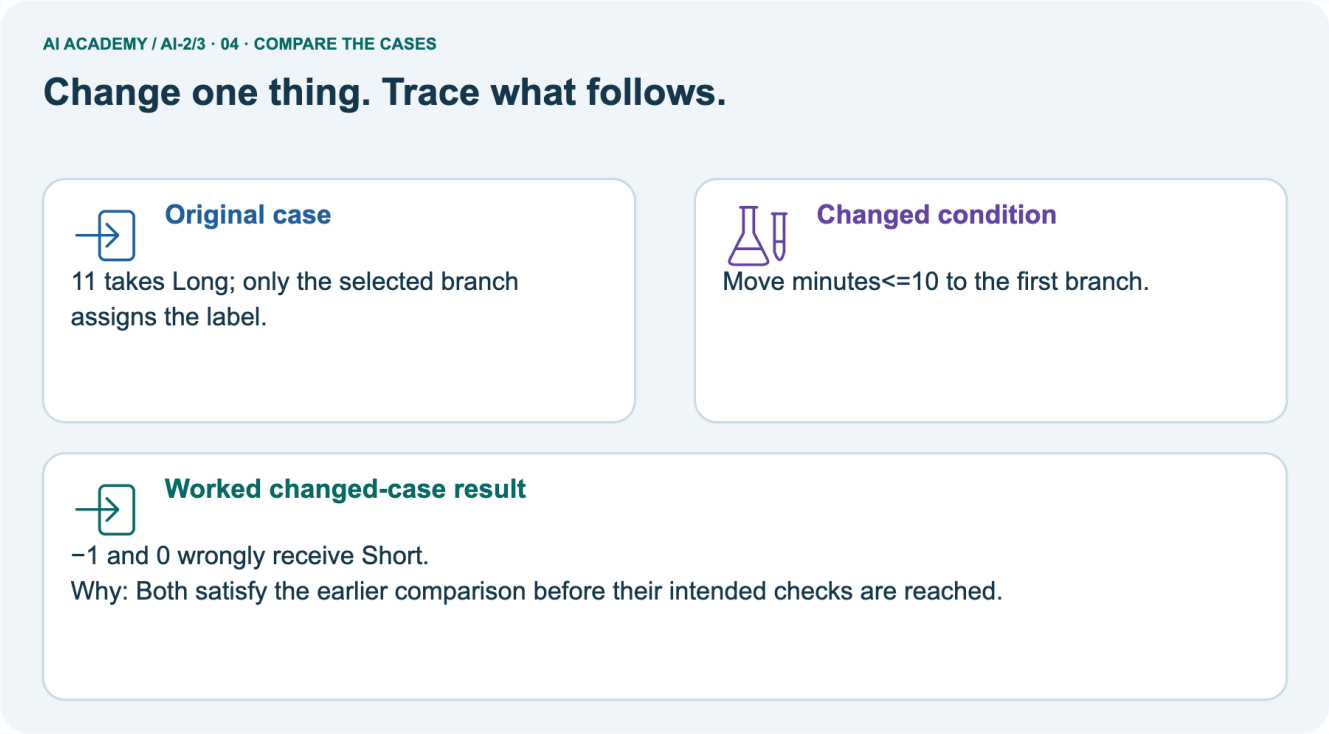
Follow the labelled picture with the matching explanation. Open the vector diagram

One chain chooses one label. -1 | Below zero is True | Invalid. 12 | Below zero False; <= 12 True | Short. 13 | Both checks False | Long

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

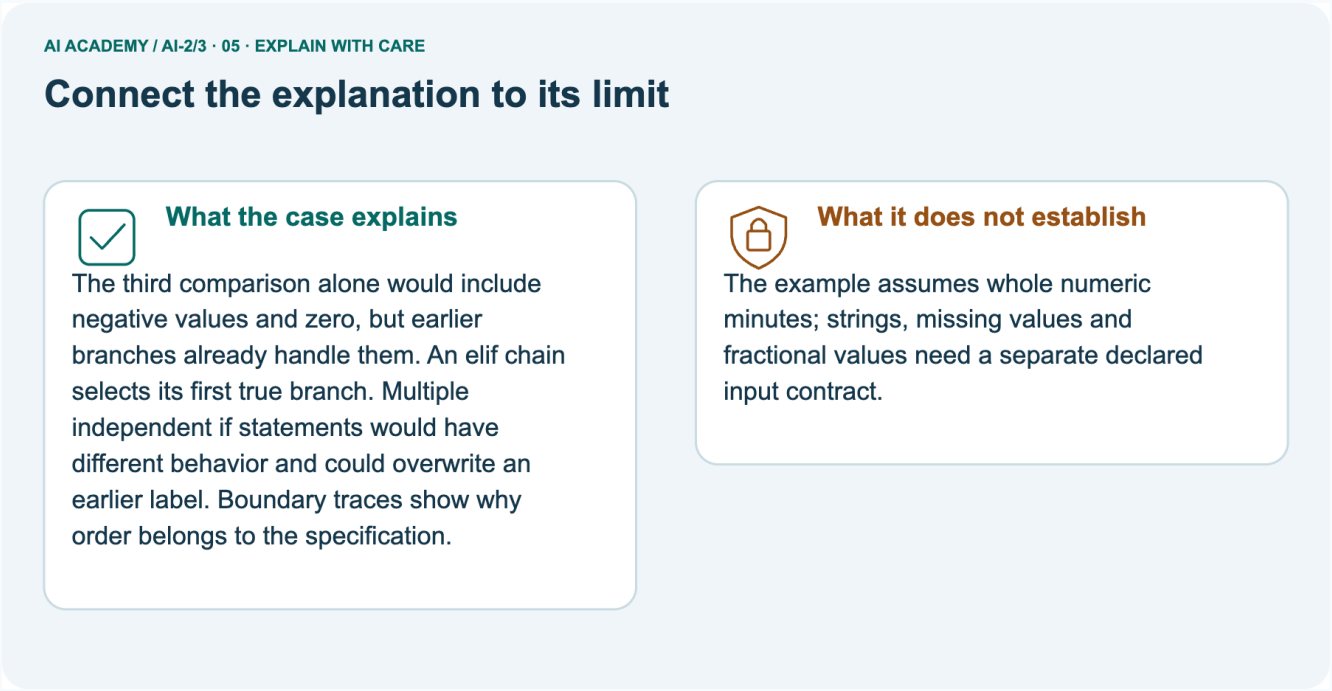
Original case: 11 takes Long; only the selected **branch** assigns the label. Changed **condition**: Move minutes<=10 to the first **branch**. Expected result: −1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

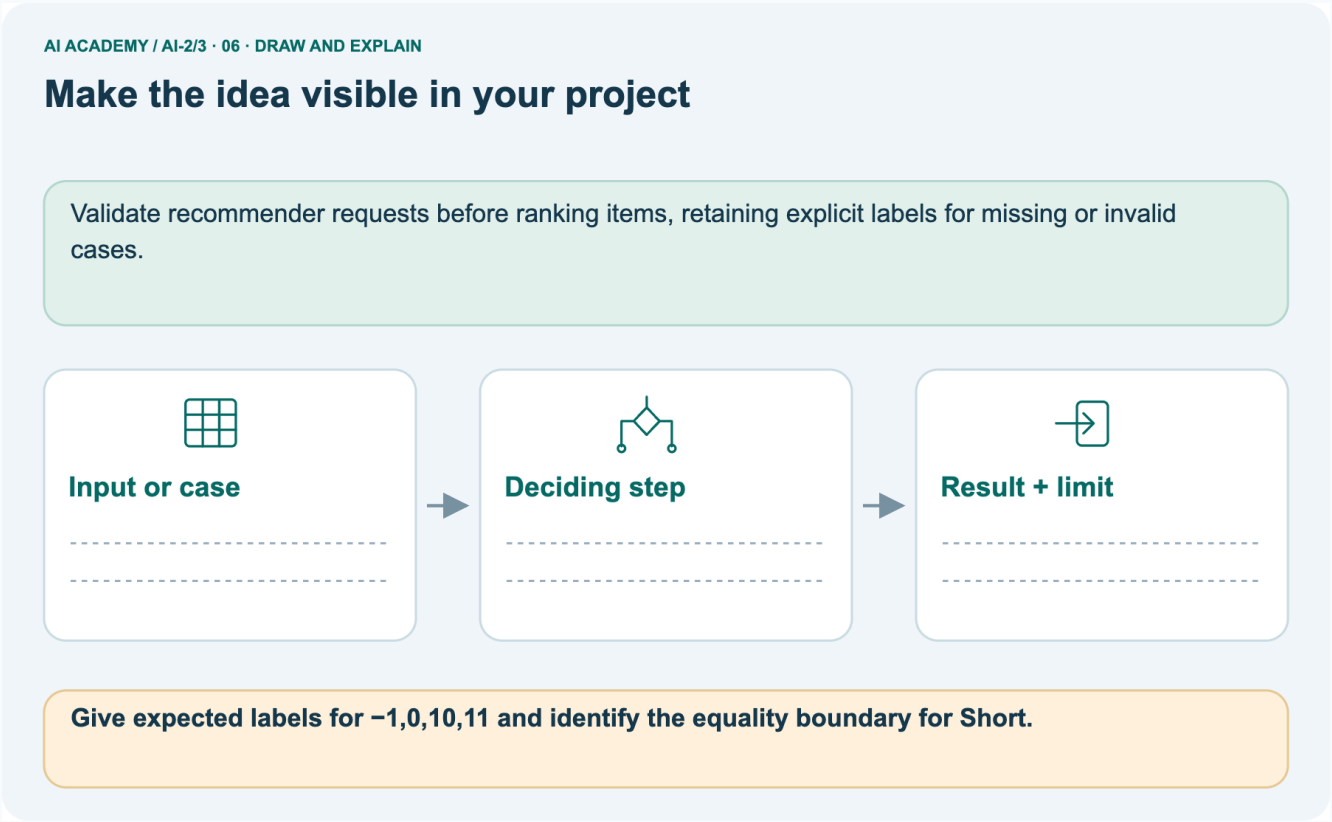
Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Limit: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases. Draw your **input**, deciding step, result and limit.

Supported question: Give expected labels for −1,0,10,11 and identify the equality **boundary** for Short.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Branching Logic: the case

02. Follow the deciding steps

03. One chain chooses one label

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.